SUSE Linux Enterprise High Performance Computing 15 SP6 (prerelease)

SUSE Linux Enterprise High Performance Computing Release Notes (Supplement to the SUSE Linux Enterprise Server Release Notes)

Starting with SLE 15 SP6, High Performance Computing is available as a module in SUSE Linux Enterprise Server. This document provides an overview of general features, capabilities, limitations, and important updates for the High Performance Computing module of SUSE Linux Enterprise Server 15 SP6 (prerelease), supplementing the main SUSE Linux Enterprise Server release notes.

This product will be released in June 2024. The latest version of these release notes is always available at https://www.suse.com/releasenotes. Drafts of the general documentation can be found at https://susedoc.github.io/doc-hpc/main/single-html/hpc-guide/.

Publication Date: 2026-08-26, Version: @VERSION@

Contents

1 About the release notes

These SUSE Linux Enterprise High Performance Computing Release Notes (Supplement to the SUSE Linux Enterprise Server Release Notes) are identical across all architectures, and the most recent version is always available online at https://www.suse.com/releasenotes.

Entries are only listed once but they can be referenced in several places if they are important and belong to more than one section.

Release notes usually only list changes that happened between two subsequent releases. Certain important entries from the release notes of previous product versions are repeated. To make these entries easier to identify, they contain a note to that effect.

However, repeated entries are provided as a courtesy only. Therefore, if you are skipping one or more service packs, check the release notes of the skipped service packs as well. If you are only reading the release notes of the current release, you could miss important changes.

2 SUSE Linux Enterprise High Performance Computing

SUSE Linux Enterprise High Performance Computing is a highly scalable, high performance open-source operating system designed to utilize the power of parallel computing for modeling, simulation and advanced analytics workloads.

SUSE Linux Enterprise High Performance Computing 15 SP6 (prerelease) provides tools and libraries related to High Performance Computing. This includes:

- Workload manager
- Remote and parallel shells
- Performance monitoring and measuring tools
- Serial console monitoring tool
- Cluster power management tool
- A tool for discovering the machine hardware topology
- System monitoring
- A tool for monitoring memory errors

- A tool for determining the CPU model and its capabilities (x86-64 only)
- User-extensible heap manager capable of distinguishing between different kinds of memory (x86-64 only)
- Serial and parallel computational libraries providing the common standards BLAS, LAPACK, …
- Various MPI implementations
- Serial and parallel libraries for the HDF5 file format

2.1 Hardware Platform Support

SUSE Linux Enterprise High Performance Computing 15 SP6 (prerelease) is available for the Intel 64/AMD64 (x86-64) and AArch64 platforms.

2.2 Important Sections of This Document

If you are upgrading from a previous SUSE Linux Enterprise High Performance Computing release, you should review at least the following sections:

- *Section 2.4, "Support statement for SUSE Linux Enterprise High Performance Computing"*

2.3 Support and life cycle

SUSE Linux Enterprise High Performance Computing is backed by award-winning support from SUSE, an established technology leader with a proven history of delivering enterprise-quality support services.

SUSE Linux Enterprise High Performance Computing 15 has a 13-year life cycle, with 10 years of General Support and 3 years of Extended Support. The current version (SP6) will be fully maintained and supported until 6 months after the release of SUSE Linux Enterprise High Performance Computing 15 SP7.

Any release package is fully maintained and supported until the availability of the next release.

Extended Service Pack Overlay Support (ESPOS) and Long Term Service Pack Support (LTSS) are also available for this product. If you need additional time to design, validate and test your upgrade plans, Long Term Service Pack Support (LTSS) can extend the support you get by an additional 12 to 36 months in 12-month increments, providing a total of 3 to 5 years of support on any given Service Pack.

For more information, see:

- The support policy at https://www.suse.com/support/policy.html
- Long Term Service Pack Support page at https://www.suse.com/support/programs/long-term-service-pack-support.html

2.4 Support statement for SUSE Linux Enterprise High Performance Computing

To receive support, you need an appropriate subscription with SUSE. For more information, see https://www.suse.com/support/programs/subscriptions/?id=SUSE_Linux_Enterprise_Server.

The following definitions apply:

L1

Problem determination, which means technical support designed to provide compatibility information, usage support, ongoing maintenance, information gathering and basic troubleshooting using available documentation.

L2

Problem isolation, which means technical support designed to analyze data, reproduce customer problems, isolate problem area and provide a resolution for problems not resolved by Level 1 or prepare for Level 3.

L3

Problem resolution, which means technical support designed to resolve problems by engaging engineering to resolve product defects which have been identified by Level 2 Support.

For contracted customers and partners, SUSE Linux Enterprise High Performance Computing is delivered with L3 support for all packages, except for the following:

- Technology Previews, see *Section 4, "Technology previews"*
- Sound, graphics, fonts and artwork
- Packages that require an additional customer contract, see *Section 2.4.1, "Software requiring specific contracts"*

SUSE will only support the usage of original packages. That is, packages that are unchanged and not recompiled.

2.4.1 Software requiring specific contracts

Certain software delivered as part of SUSE Linux Enterprise High Performance Computing may require an external contract. Check the support status of individual packages using the RPM metadata that can be viewed with `rpm`, `zypper`, or YaST.

2.4.2 Software under GNU AGPL

SUSE Linux Enterprise High Performance Computing 15 SP6 (prerelease) (and the SUSE Linux Enterprise modules) includes the following software that is shipped *only* under a GNU AGPL software license:

- Ghostscript (including subpackages)

SUSE Linux Enterprise High Performance Computing 15 SP6 (prerelease) (and the SUSE Linux Enterprise modules) includes the following software that is shipped under multiple licenses that include a GNU AGPL software license:

- MySpell dictionaries and LightProof
- ArgyllCMS

2.5 Documentation and other information

2.5.1 Available on the product media

- Read the READMEs on the media.
- Get the detailed change log information about a particular package from the RPM (where *`FILENAME`*`.rpm` is the name of the RPM):

```
rpm --changelog -qp FILENAME.rpm
```

- Check the `ChangeLog` file in the top level of the installation medium for a chronological log of all changes made to the updated packages.
- Find more information in the `docu` directory of the installation medium of SUSE Linux Enterprise High Performance Computing 15 SP6 (prerelease). This directory includes PDF versions of the SUSE Linux Enterprise High Performance Computing 15 SP6 (prerelease) Installation Quick Start Guide.

2.5.2 Online documentation

- For the most up-to-date version of the documentation for SUSE Linux Enterprise High Performance Computing 15 SP6 (prerelease), see https://susedoc.github.io/doc-hpc/main/single-html/hpc-guide/ .
- Find a collection of White Papers in the SUSE Linux Enterprise High Performance Computing Resource Library at https://www.suse.com/products/server#resources .

3 Modules, extensions, and related products

This section comprises information about modules and extensions for SUSE Linux Enterprise High Performance Computing 15 SP6 (prerelease) Modules and extensions add functionality to the system.

3.1 Modules in the SLE 15 SP6 (prerelease) product line

The SLE 15 SP6 (prerelease) product line is made up of modules that contain software packages. Each module has a clearly defined scope. Modules differ in their life cycles and update timelines.

The modules available within the product line based on SUSE Linux Enterprise 15 SP6 (prerelease) at the release of SUSE Linux Enterprise High Performance Computing 15 SP6 (prerelease) are listed in the *Modules and Extensions Quick Start* at https://documentation.suse.com/sles/15-SP3/html/SLES-all/article-modules.html .

Not all SLE modules are available with a subscription for SUSE Linux Enterprise High Performance Computing 15 SP6 (prerelease) itself (see the column *Available for*).

For information about the availability of individual packages within modules, see https://scc.suse.com/packages .

3.2 Available extensions

The following extension is not covered by SUSE support agreements, available at no additional cost and without an extra registration key: SUSE Package Hub, see https://packagehub.suse.com/ .

3.3 Related products

This sections lists related products. Usually, these products have their own release notes documents that are available from https://www.suse.com/releasenotes.

- SUSE Linux Enterprise Server: https://www.suse.com/products/server
- SUSE Linux Enterprise JeOS: https://www.suse.com/products/server/jeos
- SUSE Linux Enterprise Desktop: https://www.suse.com/products/desktop
- SUSE Linux Enterprise Server for SAP Applications: https://www.suse.com/products/sles-for-sap
- SUSE Linux Enterprise Real Time: https://www.suse.com/products/realtime
- SUSE Manager: https://www.suse.com/products/suse-manager

4 Technology previews

Technology previews are packages, stacks, or features delivered by SUSE which are not supported. They may be functionally incomplete, unstable or in other ways not suitable for production use. They are included for your convenience and give you a chance to test new technologies within an enterprise environment.

Whether a technology preview becomes a fully supported technology later depends on customer and market feedback. Technology previews can be dropped at any time and SUSE does not commit to providing a supported version of such technologies in the future.

Give your SUSE representative feedback about technology previews, including your experience and use case.

4.1 64K page size kernel flavor has been added

SUSE Linux Enterprise High Performance Computing for Arm 12 SP2 and later kernels have used a page size of 4K. This offers the widest compatibility also for small systems with little RAM, allowing to use Transparent Huge Pages (THP) where large pages make sense.

As a technology preview, SUSE Linux Enterprise High Performance Computing for Arm 15 SP6 (prerelease) adds a kernel flavor `64kb`, offering a page size of 64 KiB and physical/virtual address size of 52 bits. Same as the `default` kernel flavor, it does not use preemption.

Main purpose at this time is to allow for side-by-side benchmarking for High Performance Computing, Machine Learning and other Big Data use cases. Contact your SUSE representative if you notice performance gains for your specific workloads.

Important: Swap needs to be re-initialized

After booting the 64K kernel, any swap partitions need to re-initialized to be usable. To do this, run the `swapon` command with the `--fixpgsz` parameter on the swap partition. Note that this process deletes data present in the swap partition (for example, suspend data). In this example, the swap partition is on `/dev/sdc1`:

```
swapon --fixpgsz /dev/sdc1
```

Important: Btrfs file system uses page size as block size

It is currently not possible to use Btrfs file systems across page sizes. Block sizes below page size are not yet supported and block sizes above page size might never be supported.

During installation, change the default partitioning proposal and choose another file system, such as Ext4 or XFS, to allow rebooting from the default 4K page size kernel of the Installer into `kernel-64kb` and back.

See the *Storage Guide* for a discussion of supported file systems.

Warning: RAID 5 uses page size as stripe size

It is currently not yet possible to configure stripe size on volume creation. This will lead to sub-optimal performance if page size and block size differ.

Avoid RAID 5 volumes when benchmarking 64K vs. 4K page size kernels.

See the *Storage Guide* for more information on software RAID.

Note: Cross-architecture compatibility considerations

The SUSE Linux Enterprise High Performance Computing 15 SP6 (prerelease) kernels on x86-64 use 4K page size.

The SUSE Linux Enterprise High Performance Computing for POWER 15 SP6 (prerelease) kernel uses 64K page size.

5 Modules

5.1 HPC module

The HPC module contains HPC specific packages. These include the workload manager Slurm, the node deployment tool `clustduct`, `munge` for user authentication, the remote shell `mrsh`, the parallel shell `pdsh`, as well as numerous HPC libraries and frameworks.

It can be added or removed using the YaST UI or the `SUSEConnect` CLI tool. Refer to the system administration guide for further details.

5.2 NVIDIA Compute Module

The NVIDIA Compute Module provides the NVIDIA CUDA repository for SUSE Linux Enterprise 15. Note that that any software within this repository is under a 3rd party EULA. For more information check https://docs.nvidia.com/cuda/eula/index.html.

This module is not selected for addition by default when installing SUSE Linux Enterprise High Performance Computing. It may be selected manually during installation from the *Extension and Modules* screen. You may also select it on an installed system using YaST. To do so, run from a shell as root `yast registration`, select: `Select Extensions` and search for `NVIDIA Compute Module` and press `Next`.

> **Important**
>
> Do not attempt to add this module with the `SUSEConnect` CLI tool. This tool is not yet capable of handling 3rd party repositories.

Once you have selected this module you will be asked to confirm the 3rd party license and verify the repository signing key.

6 Changes affecting all architectures

Information in this section applies to all architectures supported by SUSE Linux Enterprise High Performance Computing 15 SP6 (prerelease).

6.1 SLE HPC no longer a separate product

As of 15 SP6 (prerelease), SUSE Linux Enterprise High Performance Computing is no longer a separate product. As a result:

- the HPC Module can now be enabled in SUSE Linux Enterprise Server
- when migrating from SUSE Linux Enterprise High Performance Computing 15 SP3, SP4, and SP5, only SUSE Linux Enterprise Server 15 SP6 will be available as migration target. The result of such a migration will be an installation of SUSE Linux Enterprise Server with all the previously enabled modules.

6.2 Enriched system visibility in the SUSE Customer Center (SCC)

SUSE is committed to helping provide better insights into the consumption of SUSE subscriptions regardless of where they are running or how they are managed; physical or virtual, on-prem or in the cloud, connected to SCC or Repository Mirroring Tool (RMT), or managed by SUSE Manager. To help you identify or filter out systems in SCC that are no longer running or decommissioned, SUSEConnect now features a daily “ping”, which will update system information automatically.

For more details see the documentation at https://documentation.suse.com/subscription/suseconnect/single-html/SLE-suseconnect-visibility/ ↗.

6.3 Automatically opened ports

Installing the following packages automatically opens the following ports:

- `dolly` - TCP ports 9997 and 9998
- `slurm` - TCP ports 6817, 6818, and 6819

Important

These release notes only document changes in SUSE Linux Enterprise High Performance Computing compared to the immediate previous service pack of SUSE Linux Enterprise High Performance Computing. The full changes and fixes can be found on the respective web site of the packages.

6.4 GNU compiler suite version 12

SLE HPC 15 SP6 (prerelease) now supports the GNU compiler suite version 12. To install the runtime environments (environment modules) for version 12, run: `zypper install gnu12-compilers-hpc`. To install all the packages required for development (C, C and Fortran compilers), run `zypper install gnu12-compilers-hpc-devel`. To load the environment, run 'module load gnu/12' in your shell. When the `-devel` package is installed, the compilers (`gcc-12`, `g-12`, `gfortran-12`) will become available in this shell under their standard names (`gcc`, `g++`, `gfortran`).

6.5 conman

`conman` has been updated to version 0.3.1:

- Fixed username/password use in `libipmiconsole.conf`.
- Added `-T` command-line option to specify terminal emulator.
- General move of files from `/usr/lib/conman` to `/usr/share/conman`.

6.6 dolly

`dolly` has been updated to version 0.64.2. The tool is less verbose by default and the `dolly` service can be activated through a socket.

6.7 imb

`imb` is shipped in version 2021.3:

- Change default value for `mem_alloc_type` to device.
- Added new IMB-MPI1-GPU benchmarks as technical preview.
- Added `-msg_pause` option.
- Changed default `window_size` from `64` to `256`.
- Added `-window_size` option for IMB-MPI1.

6.8 `memkind`

`memkind` has been updated to version 1.14.0. The full list of changes is available at http://memkind.github.io/memkind/.

6.9 `openblas`

`openblas` has been updated to version 0.3.21. It contains performance regression fixes and optimizations. For more information see https://github.com/xianyi/OpenBLAS/releases/tag/v0.3.21.

6.10 `munge`

`munge` has been updated to version 0.5.15:

- Fixed systemd service unit configuration to wait until network is online.
- Fixed sending repeated SIGTERMs to signal stop.

6.11 `cpuid`

`cpuid` has been updated to version 20221201.

This update includes:

- Added and updated identification of many CPU models and variants.
- Updated hypervisor support.
- Improved synth information and u-architecture decoding.

6.12 `lmod`

`lmod` has been updated to version 8.7.17.

The user visible changes include:

- Add option `--miniConfig` to report configuration differences from default.
- Move cache file location from `~/.lmod.d/.cache/*` to `~/.cache/lmod/*`
- Transitional support for using `~/.config/lmod` for collections. Currently collect are written to both `~/.lmod.d/` and `~/.config/lmod`.

- setenv and pushenv change local environment when running spider (and avail).
- Allow bash users to export `SUPPORT_KSH=no` so that they can avoid bash startup setting FPATH
- Add `--location` option to show to write to stderr the file location.
- Only rebuild spider caches if there are any loaded or pending modules. `module avail <name1> <name2> …` now only prints matching aliases. Search names are resolved.
- Print `dataT` table when there is an Exception.
- New command added: `module overview`.
- Add `spiderPathFilter` hook so that sites can control what paths are kept or ignored.
- Added `$LMOD_SITE_MODULEPATH` support to prepend to `MODULEPATH`
- Add support for `sh_to_modulefile` to support zsh, ksh, bash and tcsh with aliases and shell functions
- Support for source_sh added. Now support more than one shell script per modulefile.

6.13 PAPI

`PAPI` has been updated to version 7.0.0.

The highlights include:

- Added "intel_gpu" component with monitoring capabilities support for Intel GPUs, including GPU hardware events and memory performance metrics.
- Added "sysdetect" component for detecting a machine's architectural details. Additionally, PAPI offers a new API that enables users to get "sysdetect" details from within their application.
- A major redesign of the "rocm" component for advanced monitoring features for the latest AMD GPUs. The PAPI "rocm" component is now thread-safe.
- Support for NVIDIA compute capability 7.0 and greater. This implies support for CUPTI's new Profiling and Perfworks APIs.
- Significant redesign of the "sde" component into two separate entities:
    1. a standalone library "libsde" with a new API for software developers to define software-based metrics from within their applications
    2. the PAPI "sde" component that enables monitoring of these new software-based events.

- New C++ interface for "libsde," which enables software developers to define software-defined events from within their C++ applications.
- New Counter Analysis Toolkit (CAT) benchmarks and refinements of PAPI’s CAT data analysis.
- Support for FUGAKU’s A64FX Arm architecture, including monitoring capabilities for memory bandwidth and other node-wide metrics. For futher details check https://bitbucket.org/icl/papi/wiki/PAPI-Releases.md

6.14 `warewulf4`

`warewulf4` is a popular SLE HPC deployment tool whose latest version is a full rewrite in the Go programming language. It is applying lessons learned from its predecessors. It deploys minimal images which it obtains from container images stored in a registry and performs a minimal configuration for the image to be useful as a compute node image in a cluster. `warewulf4` is deprecating the former deployment tool `clustduct`.

- Update to Warewulf version 4.6.x with the following changes:
  - Default values in `nodes.conf`:
    Warewulf 4.6.x removes implicit default values for the following configuration options:
    - Kernel command line
    - Initialization parameters
    - Root
      These values are now defined in the `default` profile. Include this profile in all other profiles. The installation process automatically upgrades `nodes.conf` to update the database.
  - Split overlays:
    The overlays `wwinit` and `runtime` are now split into separate, function-specific overlays. The upgrade process automatically updates the node database. This process replaces `wwinit` and `runtime` with a list of overlays that perform the same functions.

- Site-specific and distribution overlays:

  Do not modify overlays in `/var/lib/warewulf/overlays`. Store site-specific overlays in `/etc/warewulf/overlays` instead. During an upgrade, modified overlays are saved with the `.rpmsave` suffix and moved to `/etc/warewulf/overlays/OVERLAY_NAME`.

- Package split:

  The `warewulf` suite is now split into the following packages:

  `warewulf4`
  : The base package. It contains the main binary and configuration files.

  `warewulf4-overlays`
  : Contains the overlay files to configure the Warewulf host and nodes. These files are required.

  `warewulf4-man`
  : Contains the manual pages.

  `warewulf4-dracut`
  : Required on the node image only when using `dracut` during deployment. Default installations do not require this package.

  `warewulf4-reference-doc`
  : Contains the documentation as a PDF document.

  `warewulf4-overlays-slurm`
  : Contains an overlay to simplify the installation of the Slurm workload manager.

- Update to version 4.5.8:

- Warewulf v4.5.8 simplifies the `wwinit` boot process for SELinux and configures tmpfs to spread the node image across all available NUMA nodes. It also improves the detection of kernels in the container image to more reliably detect the newest available kernel and to avoid debug / rescue kernels.

- Warewulf v4.5.7 fixes the ability to override overlay files configured in profiles with overlays configured per-node; fixes a template processing bug bug in development-time overlay rendering; and improves the preview dracut-based boot process to better support a “secure” boot process.

- added option which allows to copy in file on wwctl container exec and keep them, if they were modified.
- Update to version 4.5.6 with following changes:
  - Show more information during `wwctl container <shell|exec>` about when and if the container image will be rebuilt.
  - Command-line completion for `wwctl overlay <edit|delete|chmod|chown>`.
  - Display an error during boot if no container is defined.
  - `wwctl container list --kernel` shows the kernel detected for each container.
  - `wwctl container list --size` shows the uncompressed size of each container. `--compressed` shows the compressed size, and `--chroot` shows the size of the container source on the server.
  - Add a logrotate config for `warewulfd.log`.

6.15 Creating containers from current HPC environment

Usually users use environment modules to adjust their environment (that is, environment variables like `PATH`, `LD_LIBRARY_PATH`, `MANPATH` etc.) to pick exactly the tools and libraries they need for their work. The same can be achieved with containers by including only those components in a container that are part of this environment. This functionality is now provided using the `spack` and `singularity` applications.

6.16 Spack

6.16.1 v0.23.1

- Fix a correctness issue of `ArchSpec.intersects`.
- Make extra_attributes order independent in Spec hashing.
- Fix issue where system proxy settings were not respected in OCI build caches.
- Fix an issue where the `--test` concretizer flag was not forwarded correctly.

- Fix an issue where `codesign` and `install_name_tool` would not preserve hardlinks on Darwin.
- Fix an issue on Darwin where codesign would run on unmodified binaries.
- Patch configure scripts generated with libtool < 2.5.4, to avoid redundant flags when creating shared libraries on Darwin.
- Ensure proper UTF-8 encoding/decoding in logging.
- Fix issues related `to filter_file`.
- Fix an issue related to creating bootstrap source mirrors.
- Fix an issue where command line config arguments were not always top level.
- Fix an incorrect typehint of `concretized()`.
- Improve mention of next Spack version in warning. Tests: fix forward compatibility with Python 3.13.
- Docs: encourage use of `--oci-username-variable` and `--oci-password-variable`.
- Docs: ensure Getting Started has bootstrap list output in correct place.

6.16.2 v0.23.0

This is a new major release.

6.16.2.1 Features in this Release

- Spec splicing To make binary installation more seamless in Spack, `v0.23` introduces "splicing", which allows users to deploy binaries using local, optimized versions of a binary interface, even if they were not built with that interface. For example, this would allow you to build binaries in the cloud using `mpich` and install them on a system using a local, optimized version of `mvapich2` *without rebuilding*. Spack preserves full provenance for the installed packages and knows that they were built one way but deployed another. The intent is to leverage this across many key HPC binary packages, e.g. MPI, CUDA, ROCm, and libfabric. Fundamentally, splicing allows Spack to redeploy an existing spec with different dependencies than how it was built. There are two interfaces to splicing.
  . Explicit Splicing In the concretizer config, you can specify a target spec and a replacement by hash.

```
concretizer:
  splice:
    explicit:
    - target: mpi
      replacement: mpich/abcdef
```

Here, every installation that would normally use the target spec will instead use its replacement. Above, any spec using *any* `mpi` will be spliced to depend on the specific `mpich` installation requested. This *can* go wrong if you try to replace something built with, e.g., `openmpi` with `mpich`, and it is on the user to ensure ABI compatibility between target and replacement specs. This currently requires some expertise to use, but it will allow users to reuse the binaries they create across more machines and environments. . Automatic Splicing (experimental) In the concretizer config, enable automatic splicing:

```
  concretizer:
    splice:
      automatic: true
```

or run:

```
spack config add concretizer:splice:automatic:true
```

The concretizer will select splices for ABI compatibility to maximize package reuse. Packages can denote ABI compatibility using the `can_splice` directive. No packages in Spack yet use this directive, so if you want to use this feature you will need to add `can_splice` annotations to your packages. We are working on ways to add more ABI compatibility information to the Spack package repository, and this directive may change in the future.
Further documentation: **https://spack.readthedocs.io/en/latest/build_settings.html#splicing** https://spack.readthedocs.io/en/latest/packaging_guide.html#specifying-abi-compatibility

- Broader variant propagation You can specify propagated variants like `hdf5 build_type==RelWithDebInfo` or `trilinos ++openmp` to propagate a variant to all dependencies for which it is relevant. This is valid *even* if the variant does not exist on the package or its dependencies. See https://spack.readthedocs.io/en/latest/basic_usage.html#variants.

- Query specs by namespace Allow a package's namespace (indicating the repository it came from) to be treated like a variant. You can request packages from particular repos like this:

```
spack find zlib namespace=builtin
```

```
spack find zlib namespace=myrepo
```

Previously, the spec syntax only allowed namespaces to be prefixes of spec names, e.g. `builtin.zlib`. The previous syntax still works.

- `spack spec` respects environment settings and `unify:true` `spack spec` did not previously respect environment lockfiles or unification settings, which made it difficult to see exactly how a spec would concretize within an environment. Now it does, so the output you get with `spack spec` will be *the same* as what your environment will concretize to when you run `spack concretize`. Similarly, if you provide multiple specs on the command line with `spack spec`, it will concretize them together if `unify:true` is set.
- Less noisy `spack spec` output `spack spec` previously showed output like this:

  ```
  > spack spec /v5fn6xo
  Input spec
  --------------------------------
   -   /v5fn6xo

   Concretized
  --------------------------------
  [+]  openssl@3.3.1%apple-clang@16.0.0~docs+shared arch=darwin-sequoia-m1
  ...
  ```

  But the input spec is redundant, and we know we run `spack spec` to concretize the input spec. `spack spec` now *only* shows the concretized spec.
- Better output for `spack find -c` In an environment, `spack find -c` lets you search the concretized, but not yet installed, specs, just as you would the installed ones. As with `spack spec`, this should make it easier for you to see what *will* be built before building and installing it.
- `spack -C <env>`: use an environment's configuration without activation Spack environments allow you to associate:

  1. a set of (possibly concretized) specs, and
  2. configuration When you activate an environment, you're using both of these.
     Previously, we supported:

     - `spack -e <env>` to run spack in the context of a specific environment, and
     - `spack -C <directory>` to run spack using a directory with configuration files.

You can now also pass an environment to `spack -C` to use *only* the environment's configuration, but not the specs or lockfile.

6.16.2.2 New commands, options, and directives

- The new `spack env track` command takes a non-managed Spack environment and adds a symlink to Spack's `$environments_root` directory, so that it will be included for reference counting for commands like `spack uninstall` and `spack gc`. If you use free-standing directory environments, this is useful for preventing Spack from removing things required by your environments. You can undo this tracking with the `spack env untrack` command.
- Add `-t` short option for `spack --backtrace` `spack -d / --debug` enables backtraces on error, but it can be very verbose, and sometimes you just want the backtrace. `spack -t / --backtrace` provides that option.
- `gc`: restrict to specific specs If you only want to garbage-collect specific packages, you can now provide them on the command line. This gives users finer-grained control over what is uninstalled.
- oci buildcaches now support `--only=package`. You can now push *just* a package and not its dependencies to an OCI registry. This allows dependents of non-redistributable specs to be stored in OCI registries without an error.

6.16.2.3 Highlighted bugfixes

- Externals no longer override the preferred provider. External definitions could interfere with package preferences. Now, if `openmpi` is the preferred `mpi`, and an external `mpich` is defined, a new `openmpi` *will* be built if building it is possible. Previously we would prefer `mpich` despite the preference.
- Composable `cflags`. This release fixes a longstanding bug that concretization would fail if there were different `cflags` specified in `packages.yaml`, `compilers.yaml`, or on `the` CLI. Flags and their ordering are now tracked in the concretizer and flags from multiple sources will be merged.
- Fix concretizer Unification for included environments.

6.16.2.4 Deprecations, removals, and syntax changes

- The old concretizer has been removed from Spack, along with the `config:concretizer` config option. Spack will emit a warning if the option is present in user configuration, since it now has no effect. Spack now uses a simpler bootstrapping mechanism, where a JSON prototype is tweaked slightly to get an initial concrete spec to download.
- Best-effort expansion of spec matrices has been removed. This feature did not work with the "new" ASP-based concretizer, and did not work with `unify: True` or `unify: when_possible`. Use the exclude key (https://spack.readthedocs.io/en/latest/environments.html#spec-matrices) for the environment to exclude invalid components, or use multiple spec matrices to combine the list of specs for which the constraint is valid and the list of specs for which it is not.
- The old Cray `platform` (based on Cray PE modules) has been removed, and `platform=cray` is no longer supported. Since `v0.19`, Spack has handled Cray machines like Linux clusters with extra packages, and we have encouraged using this option to support Cray. The new approach allows us to correctly handle Cray machines with non-SLES operating systems, and it is much more reliable than making assumptions about Cray modules. See the `v0.19` release notes for more details.
- The `config:install_missing_compilers` config option has been deprecated, and it is a no-op when set in `v0.23`. Our new compiler dependency model will replace it with a much more reliable and robust mechanism in `v1.0`.
- Config options that deprecated in `v0.21` have been removed in `v0.23`. You can now only specify preferences for `compilers`, `targets`, and `providers` globally via the `packages:all:` section. Similarly, you can only specify `versions:` locally for a specific package.
- Spack's old test interface has been removed, having been deprecated in `v0.22.0`. All `builtin` packages have been updated to use the new interface. See the stand-alone test documentation (https://spack.readthedocs.io/en/latest/packaging_guide.html#stand-alone-tests)
- The `spack versions --safe-only` option, deprecated since `v0.21.0`, has been removed.
- The `--dependencies` and `--optimize` arguments to `spack ci` have been deprecated.

6.16.2.5 Binary caches

- Public binary caches now include an ML stack for Linux/aarch64. We now build an ML stack for Linux/aarch64 for all pull requests and on develop. The ML stack includes both CPU-only and CUDA builds for Horovod, Hugging Face, JAX, Keras, PyTorch, scikit-learn, TensorBoard, and TensorFlow, and related packages. The CPU-only stack also includes XGBoost. See https://cache.spack.io/tag/develop/?stack=ml-linux-aarch64-cuda ↗.

6.16.2.6 Architecture support

- archspec has been updated to `v0.2.5`, with support for `zen5`.
- Spack's CUDA package now supports the Grace Hopper `9.0a` compute capability.

6.16.2.7 Other notable changes

- Bugfix: `spack find -x` in environments.
- Spec splices are now robust to duplicate nodes with the same name in a spec.
- Cache per-compiler libc calculations for performance.
- Fixed a bug in external detection for openmpi.
- Mirror configuration allows username/password as environment variables.
- Default library search caps maximum depth/
- Unify interface for `spack spec` and `spack solve` commands.
- Spack no longer RPATHs directories in the default library search path.
- Improved performance of Spack database.
- Enable package reuse for packages with versions from git refs.
- Improved tracking of task queueing/requeueing in the installer.

6.16.3 v0.22.2

6.16.3.1 Bugfixes

- Forward compatibility with Spack 0.23 packages with language dependencies.
- Forward compatibility with `urllib` from Python 3.12.6+.
- Bump vendored `archspec` for better aarch64 support.
- Fix regression in `\\{variants.X}` and `\\{variants.X.value}` format strings.
- Ensure shell escaping of environment variable values in load and activate commands.
- Fix an issue where `spec[pkg]` considers specs outside the current DAG.
- Do not halt concretization on unknown variants in externals.
- Improve validation of `develop` config section/
- Explicitly disable `ccache` if turned off in config, to avoid cache pollution.
- Improve backwards compatibility in `include_concrete`.
- Fix issue where package tags were sometimes repeated.
- Make `setup-env.sh` “sourced only” by dropping execution bits.
- Make certain source/binary fetch errors recoverable instead of a hard error.
- Remove debug statements in package hash computation.
- Remove redundant clingo warnings.
- Remove hard-coded layout version.
- Do not initialize previous store state in `use_store`.

6.16.3.2 Package updates

- `chapel` major update/

6.16.4 v0.22.1

6.16.4.1 Bug Fixes

- Fix reuse of externals on Linux.
- Ensure parent gcc-runtime version >= child.
- Ensure the latest gcc-runtime is rpath'ed when multiple exist among link deps.
- Improve version detection of glibc.
- Improve heuristics for solver.
- Make strong preferences override reuse.
- Reduce verbosity when C compiler is missing.
- Make missing ccache executable an error when required.
- Make every environment view containing `python` a `venv`.
- Fix external detection for compilers with os but no target.
- Fix version optimization for roots.
- Handle common implementations of pagination of tags in OCI build caches.
- Apply fetched patches to develop specs.
- Avoid Windows wrappers for filesystem utilities on non-Windows.
- Fix formatting issue in `spack audit`.

6.16.4.2 Package updates

- Require libiconv for iconv. Notice that glibc/musl also provide iconv, but are not guaranteed to be complete. Set `packages:iconv:require:[glibc]` to restore the old behavior.
- protobuf: fix 3.4:3.21 patch checksum.

- protobuf: update hash for patch needed when="@3.4:3.21".
- git: bump v2.39 to 2.45; deprecate unsafe versions.
- gcc: use `-rpath \{rpath_dir}` not `-rpath=\\{rpath_dir}`.
- Remove mesa18 and libosmesa.
- Enforce consistency of `gl` providers.
- py-matplotlib: qualify when to do a post install.
- rust: fix v1.78.0 instructions.
- suite-sparse: improve setting of the libs property.
- netlib-lapack: provide blas and lapack together.

6.17 Slurm

6.17.1 Deprecation of old Versions of Slurm

SLE receives a new Slurm version for roughly every second upstream release. To facilitate this, old version of Slurm will go out of maintenance successively. Users of these versions are encouraged to migrate to a later version before the expiry date. Note, that Slurm only allows migration two versions upwards without loss of data. To migrate to the latest version, migrations to intermedite versions may be required.

Once Slurm versions are out of maintenance, updates to to packages depending on this Slurm version will no longer be provided. In particular, this will be the case for `pdsh-slurm` - the Slurm plugin to Pdsh. Note that `pdsh-slurm` is compatible to the Slurm initially shipped with a product/service pack) while packages `pdsh-slurm_<slurm_version>` is compatible with an upgade version of Slurm.

TABLE 1: TABLE SUNSET SCHEDULE FOR SLURM

Slurm Version	Released for Service Pack	Support End Date
17.02	SLE-12-SP2	May 2024
18.08	SLE-15-SP1 and older	October 2024
20.02	SLE-15-SP2 and older	Januar 2025
20.11	SLE-15-SP3/4 and older	January 2026

22.05	SLE-15-SP4 and older	December 2026
23.02	SLE-15-SP5/6 and older	January 2028

6.17.2 Slurm 23.02

6.17.2.1 Important Notes on Upgrading Slurm from a Previous Version

If using the `slurmdbd` (Slurm DataBase Daemon) you must update this first.

If using a backup DBD you must start the primary first to do any database conversion, the backup will not start until this has happened.

The 23.02 `slurmdbd` will work with Slurm daemons of version 21.08 and above. You will not need to update all clusters at the same time, but it is very important to update `slurmdbd` first and having it running before updating any other clusters making use of it.

Slurm can be upgraded from version 22.05 to version 23.02 without loss of jobs or other state information. Upgrading directly from an earlier version of Slurm will result in loss of state information.

All `SPANK` plugins must be recompiled when upgrading from any Slurm version prior to 23.02.

> **Note**
>
> PMIx v1.x is no longer supported.

6.17.2.2 Highlights

- `slurmctld` - Add new RPC rate limiting feature. This is enabled through `SlurmctldParameters=rl_enable`, otherwise disabled by default.
- Make `scontrol` reconfigure and sending a `SIGHUP` to the `slurmctld` behave the same. If you were using `SIGHUP` as a 'lighter' scontrol reconfigure to rotate logs please update your scripts to use SIGUSR2 instead.
- Change cloud nodes to show by default. `PrivateData=cloud` is no longer needed.
- `sreport` - Count planned (FKA reserved) time for jobs running in `IGNORE_JOBS` reservations. Previously was lumped into `IDLE` time.

- `job_container/tmpfs` - Support running with an arbitrary list of private mount points (`/tmp` and `/dev/shm` are the default, but not required).
- `job_container/tmpfs` - Set more environment variables in `InitScript`.
- Make all cgroup directories created by Slurm owned by root. This was the behavior in cgroup/v2 but not in cgroup/v1 where by default the step directories ownership were set to the user and group of the job.
- `accounting_storage/mysql` - change purge/archive to calculate record ages based on end time, rather than start or submission times.
- `job_submit/lua` - add support for `log_user()` from `slurm_job_modify()`.
- Run the following scripts in slurmscriptd instead of `slurmctld`: `ResumeProgram`, `ResumeFailProgram`, `SuspendProgram`, `ResvProlog`, `ResvEpilog`, and `RebootProgram` (only with `SlurmctldParameters=reboot_from_controller`).
- Only permit changing log levels with `srun --slurmd-debug` by root or `SlurmUser`.
- `slurmctld` will `fatal()` when reconfiguring the `job_submit` plugin fails.
- Add PowerDownOnIdle partition option to power down nodes after nodes become idle.
- Add “[jobid.stepid]” prefix from `slurmstepd` and “slurmscriptd” prefix from `slurmcriptd` to Syslog logging. Previously was only happening when logging to a file.
- Add purge and archive functionality for job environment and job batch script records.
- Extend support for Include files to all "configless" client commands.
- Make node weight usable for powered down and rebooting nodes.
- Removed “launch” plugin.
- Add “Extra” field to job to store extra information other than a comment.
- Add usage gathering for AMD (requires ROCM 5.5+) and NVIDIA gpus.
- Add job’s allocated nodes, features, oversubscribe, partition, and reservation to `SLURM_RESUME_FILE` output for power saving.
- Automatically create directories for stdout/stderr output files. Paths may use `%j` and related substitution characters as well.
- Add `--tres-per-task` to `salloc`/`sbatch`/`srun`.

- Allow `nodefeatures` plugin features to work with cloud nodes. e.g. - Powered down nodes have no active changeable features.
  - Nodes can’t be changed to other active features until powered down.
  - Active changeable features are reset/cleared on power down.
- Make `slurmstepd` cgroups constrained by total configured memory from `slurm.conf` (`NodeName=<> RealMemory=#`) instead of total physical memory.
- `node_features/helpers` - add support for the OR and parentheses operators in a `--constraint` expression.
- `slurmctld` will `fatal()` when [Prolog|Epilog]Slurmctld are defined but are not executable.
- Validate node registered active features are a super set of node’s currently active changeable features.
- On clusters without any `PrologFlags` options, batch jobs with failed prologs nolonger generate an output file.
- Add `SLURM_JOB_START_TIME` and `SLURM_JOB_END_TIME` environment variables.
- Add `SuspendExcStates` option to `slurm.conf` to avoid suspending/powering down specific node states.
- Add support for DCMI power readings in IPMI plugin.
- `slurmrestd` served `/slurm/v0.0.39` and `/slurmdb/v0.0.39` endpoints had major changes from prior versions. Almost all schemas have been renamed and modified. Sites using OpenAPI Generator clients are highly suggested to upgrade to to using atleast version 6.x due to limitations with prior versions.
- Allow for `--nodelist` to contain more nodes than required by `--nodes`.
- Rename “nodes” to “nodes_resume” in `SLURM_RESUME_FILE` job output.
- Rename “all_nodes” to “all_nodes_resume” in `SLURM_RESUME_FILE` output.
- Add `jobcomp/kafka` plugin.
- Add new `PreemptParameters=reclaim_licenses` option which will allow higher priority jobs to preempt jobs to free up used licenses. (This is only enabled for with `PreemptModes` of `CANCEL` and `REQUEUE`, as Slurm cannot guarantee suspended jobs will release licenses correctly.)
- `hpe/slingshot` - add support for the instant-on feature.

- Add ability to update `SuspendExc*` parameters with `scontrol`.
- Add ability to restore `SuspendExc*` parameters on restart with `slurmctld -R` option.
- Add ability to clear a GRES specification by setting it to "0" via “scontrol update job”.
- Add `SLURM_JOB_OVERSUBSCRIBE` environment variable for `Epilog`, `Prolog`, `EpilogSlurmctld`, `PrologSlurmctld`, and mail ouput.
- System node down reasons are appended to existing reasons, separated by ':'.
- New command scrun has been added. scrun acts as an Open Container Initiative (OCI) runtime proxy to run containers seamlessly via Slurm.
- Fixed `GpuFreqDef` option. When set in `slurm.conf`, it will be used if `--gpu-freq` was not explicitly set by the job step.

6.17.2.3 Configuration File Changes (see appropriate man page for details)

- `job_container.conf` - Added “Dirs” option to list desired private mount points.
- `node_features` plugins - invalid users specified for `AllowUserBoot` will now result in `fatal()` rather than just an error.
- Deprecate `AllowedKmemSpace`, `ConstrainKmemSpace`, `MaxKmemPercent`, and `MinKmemSpace`.
- Allow jobs to queue even if the user is not in `AllowGroups` when `EnforcePartLimits=no` is set. This ensures consistency for all the Partition access controls, and matches the documented behavior for `EnforcePartLimits`.
- Add `InfluxDBTimeout` parameter to `acct_gather.conf`.
- `job_container/tmpfs` - add support for expanding `%h` and `%n` in `BasePath`.
- `slurm.conf` - Removed `SlurmctldPlugstack` option.
- Add new `SlurmctldParameters=validate_nodeaddr_threads=<number>` option to allow concurrent hostname resolution at `slurmctld` startup.
- Add new `AccountingStoreFlags=job_extra` option to store a job’s extra field in the database.
- Add new “defer_batch” option to `SchedulerParameters` to only defer scheduling for batch jobs.

- Add new `DebugFlags` option “JobComp” to replace “Elasticsearch”.
- Add configurable job requeue limit parameter - `MaxBatchRequeue` - in `slurm.conf` to permit changes from the old hard-coded value of 5.
- `helpers.conf` - Allow specification of node specific features.
- `helpers.conf` - Allow many features to one helper script.
- `job_container/tmpfs` - Add “Shared” option to support shared namespaces. This allows autofs to work with the `job_container/tmpfs` plugin when enabled.
- `acct_gather.conf` - Added `EnergyIPMIPowerSensors=Node=DCMI` and `Node=DCMI_ENHANCED`.
- Add new “getnameinfo_cache_timeout=<number>” option to CommunicationParameters to adjust or disable caching the results of `getnameinfo()`.
- Add new PrologFlags=ForceRequeueOnFail option to automatically requeue batch jobs on Prolog failures regardless of the job --requeue setting.
- Add `HealthCheckNodeState=NONDRAINED_IDLE` option.
- Add “explicit” to Flags in `gres.conf`. This makes it so the gres is not automatically added to a job’s allocation when `--exclusive` is used. Note that this is a per-node flag.
- Moved the “preempt_” options from `SchedulerParameters` to `PreemptParameters`, and dropped the prefix from the option names. (The old options will still be parsed for backwards compatibility, but are now undocumented.)
- Add `LaunchParameters=ulimit_pam_adopt`, which enables setting `RLIMIT_RSS` in adopted processes.
- Update SwitchParameters=job_vni to enable/disable creating job VNIs for all jobs, or when a user requests them.
- Update `SwitchParameters=single_node_vni` to enable/disable creating single node VNIs for all jobs, or when a user requests them.
- Add ability to preserve `SuspendExc*` parameters on reconfig with `ReconfigFlags=KeepPowerSaveSettings`.
- `slurmdbd.conf` - Add new `AllResourcesAbsolute` to force all new resources to be created with the `Absolute` flag.

- `topology/tree` - Add new `TopologyParam=SwitchAsNodeRank` option to reorder nodes based on switch layout. This can be useful if the naming convention for the nodes does not natually map to the network topology.
- Removed the default setting for `GpuFreqDef`. If unset, no attempt to change the GPU frequency will be made if `--gpu-freq` is not set for the step.

6.17.2.4 Command Changes (see man pages for details)

- `sacctmgr` - no longer force updates to the `AdminComment`, `Comment`, or `SystemComment` to lower-case.
- `sinfo` - Add `-F/--future` option to sinfo to display future nodes.
- `sacct` - Rename “Reserved” field to “Planned” to match sreport and the nomenclature of the 'Planned' node.
- `scontrol` - advanced reservation flag `MAINT` will no longer replace nodes, similar to `STATIC_ALLOC`
- `sbatch` - add parsing for `#PBS -d` and `#PBS -w`.
- `scontrol` show `assoc_mgr` will show username(uid) instead of uid in QoS section.
- Add `strigger --draining` and `-R`/`--resume` options.
- Change `--oversubscribe` and `--exclusive` to be mutually exclusive for job submission. Job submission commands will now fatal if both are set. Previously, these options would override each other, with the last one in the job submission command taking effect.
- `scontrol` - Requested `TRES` and allocated `TRES` will now always be printed when showing jobs, instead of one `TRES` output that was either the requested or allocated.
- `srun --ntasks-per-core` now applies to job and step allocations. Now, use of `--ntasks-per-core=1` implies `--cpu-bind=cores` and `--ntasks-per-core>1` implies `--cpu-bind=threads`.
- `salloc`/`sbatch`/`srun` - Check and abort if ntasks-per-core > threads-per-core.
- `scontrol` - Add `ResumeAfter=<secs>` option to “scontrol update nodename=”.
- Add a new “nodes=” argument to `scontrol setdebug` to allow the debug level on the `slurmd` processes to be temporarily altered.

- Add a new “nodes=” argument to “scontrol setdebugflags” as well.
- Make it so `scrontab` prints client-side the `job_submit()` error messsage (which can be set i.e. by using the `log_user()` function for the lua plugin).
- `scontrol` - Reservations will not be allowed to have `STATIC_ALLOC` or `MAINT` flags and `REPLACE[_DOWN]` flags simultaneously.
- `scontrol` - Reservations will only accept one reoccurring flag when being created or updated.
- `scontrol` - A reservation cannot be updated to be reoccurring if it is already a floating reservation.
- `squeue` - removed unused “%s” and “SelectJobInfo” formats.
- `squeue` - align print format for exit and derived codes with that of other components (`<exit_status>:<signal_number>`).
- `sacct` - Add `--array` option to expand job arrays and display array tasks on separate lines.
- Partial support for “--json” and “--yaml” formated outputs have been implemented for `sacctmgr`, `sdiag`, `sinfo`, `squeue`, and `scontrol`. The resultant data ouput will be filtered by normal command arguments. Formatting arguments will continue to be ignored.
- `salloc`/`sbatch`/`srun` - extended the `--nodes` syntax to allow for a list of valid node counts to be allocated to the job. This also supports a "step count" value (e.g., `--nodes=20-100:20` is equivalent to `--nodes=20,40,60,80,100`) which can simplify the syntax when the job needs to scale by a certain "chunk" size.
- `srun` - add user requestible vnis with “--network=job_vni” option.
- `srun` - add user requestible single node VNIs with the “--network=single_node_vni” option.

6.17.2.5 API Changes

- `job_container` plugins - `container_p_stepd_create()` function signature replaced `uint32_t` uid with `stepd_step_rec_t*` step.
- `gres` plugins - `gres_g_get_devices()` function signature replaced `pid_t` pid with `stepd_step_rec_t*` step.
- `cgroup` plugins - `task_cgroup_devices_constrain()` function signature removed `pid_t` pid.

- `task` plugins - replace `task_p_pre_set_affinity()`, `task_p_set_affinity()`, and `task_p_post_set_affinity()` with `task_p_pre_launch_priv()` like it was back in slurm 20.11.
- Allow for concurrent processing of `job_submit_g_submit()` and `job_submit_g_modify()` calls. If your plugin is not capable of concurrent operation you must add additional locking within your plugin.
- Removed return value from `slurm_list_append()`.
- The `List` and `ListIterator` types have been removed in favor of `list_t` and `list_itr_t` respectively.
- `burst buffer` plugins - add `bb_g_build_het_job_script()`. `bb_g_get_status()` - added authenticated UID and GID. `bb_g_run_script()` - added `job_info` argument.
- `burst_buffer.lua` - Pass UID and GID to most hooks. Pass `job_info` (detailed job information) to many hooks. See `etc/burst_buffer.lua.example` for a complete list of changes. WARNING: Backwards compatibility is broken for `slurm_bb_get_status`: UID and GID are passed before the variadic arguments. If UID and GID are not explicitly listed as arguments to `slurm_bb_get_status()`, then they will be included in the variadic arguments. Backwards compatibility is maintained for all other hooks because the new arguments are passed after the existing arguments.
- `node_features` plugins - `node_features_p_reboot_weight()` function removed. `node_features_p_job_valid()` - added parameter `feature_list`. `node_features_p_job_xlate()` - added parameters `feature_list` and `job_node_bitmap`.
- New `data_parser` interface with v0.0.39 plugin.

6.17.2.6 Known Issues

- The `--uid` option for the `srun` command is broken: its use may lead to the error message `job <job ID> queued and waiting for resources`. This does not, however, affect the `sbatch` command.

6.18 Apptainer

- Update to version 1.3.6

- Avoid using kernel overlayfs when the lower layer is a sandbox on an incompatible filesystem type such as GPFS or Lustre. For those cases use fuse-overlayfs instead. This fixes a regression introduced in 1.3.0. The regression didn't much impact Lustre because kernel overlayfs refused to try to use it and Apptainer proceeded to use fuse-overlayfs anyway, but with GPFS the kernel overlayfs allowed mounting but returned stale file handle errors.

- Version 1.3.5
  - Fix a regression introduced in 1.3.4 that overwrote existing standard `/.singularity.d` files such as `runscript` in container images even if they had been modified.
  - Skip attempting to bind inaccessible mount points when handling the `mount hostfs = yes` configuration option.
  - Support parsing nested variables defined inside `%arguments` section of definition files.
  - Ignore invalid environment variables when pulling oci/docker containers.
- Version 1.3.4
  - Fixed sif-embedded overlay partitions for containers that are larger than 2 gigabytes.
  - Fixed the failure when starting apptainer with `instance --fakeroot`.
  - `apptainer build -B ...` can now be used to mount custom resolv.conf and hosts files from non-standard outside locations. This can be used to run `apptainer build` in a nix-build sandbox that has no `/etc/resolv.conf`.
  - Fixed failing builds from local images that have symbolic links for paths that are part of the base container environment (e.g. /var/tmp → /tmp).
  - Show info messages suggesting to use `enable underlay = preferred` or the `--underlay` flag when overlay is implied for bind mounts but the kernel is too old to support fuse mounts in user namespaces and so tries to use fusermount.
  - When someone uses a `yum` bootstrap to build a container without using subuid-based fakeroot or root, warn that it is unlikely to work.
  - Allow a writable `--overlay` to be used with `--nvccli` instead of `--writable-tmpfs`.
  - If an error "no descriptor found for reference" is seen while getting an oci container, retry the operation up to five times.

- Make fakeroot Recommended for SUSE rpms instead of Required.
- Allow bind mounts onto existing files on r/o NFS filesystems.
- If an error is seen in the %post section when building a container using fakeroot mode 3 (with the fakeroot command) then show a message suggesting using `--ignore-fakeroot-command` and referring to the documentation about how to install and use it inside the container definition file.
- Show a more helpful error message when using fakeroot in suid mode and there's an `/etc/subuid` mapping even though user namespaces are not available (user namespaces are required for `/etc/subuid` mapping).

- Version 1.3.3
  - Added libcudadebugger.so to nvliblist.conf to support cuda-gdb in CUDA 12+.
  - Ensure opened/kept file descriptors in stage 1 are not closed during the Go garbage collection to avoid "bad file descriptor" errors at startup.
  - Fixed a segmentation violation issue when running Apptainer checkpoint.
  - Fixed an issue that Apptainer won't read default docker credentials.
- Version 1.3.2
  - Fix for CVE-2024-3727 (https://bugzilla.suse.com/show_bug.cgi?id=1224114) in a dependent library which describes a flaw that can allow attackers to trigger unexpected authenticated registry accesses due to object digest values not being validated in all cases.
  - Fixed the issue when nesting `apptainer instance start` inside a container on cgroups-v2 capable host.
  - Fixed the issue that oras download progress bar gets stuck when downloading large images.
- Version 1.3.1
  - Make "apptainer build" work with signed Docker containers.
  - Fixed regression introduced in 1.3.0 that prevented closing cryptsetup and the corresponding loop device after running an encrypted sif container file in suid mode.

- Stopped binding over the default timezone in the container with the host's timezone, which led to unexpected behavior if the application changed timezones.
- Added progress bars for `oras://` push and pull.
- Hide `Instance stats will not be available` message under `--sharens` mode.
- Fix problem where credentials locally stored with `registry login` command were not usable in some execution flows. Run `registry login` again with latest version to ensure credentials are stored correctly.
- Make runscript timeout configurable.
- Return invalid bind path mount options during bind path parsing.
- Make the INFO message more helpful when a running background process at exit time causes a FUSE mount to not shut down cleanly.
- Fixed the wrong mediaType in the oras push manifest.

- Use go-jose version with fix for CVE-2024-28180 (https://bugzilla.suse.com/show_bug.cgi?id=1235211).

7 Removed and deprecated features and packages

This section lists features and packages that were removed from SUSE Linux Enterprise High Performance Computing or will be removed in upcoming versions.

- `dapl`, `rds-tools`, and `imgen` are being deprecated due to lack of upstream activity.
- `openmpi` 2 and `openmpi` 3 are being deprecated due to being replaced by `openmpi` 4.

7.1 Removed features and packages

The following features and packages have been removed in this release.

- Python 2 bindings for `genders` has been removed. These are now provided for Python 3.
- Ganglia is not supported anymore in 15 SP6 (prerelease). It has been replaced with Grafana (https://grafana.com/)
- Due to a lack of usage by customers, some library packages have been removed from the HPC module in SLE HPC 15 SP6 (prerelease). On SUSE Linux Enterprise you can build your own library using `spack`. These libraries will continue to be available through SUSE Package Hub. The following libraries have been removed:
  - `boost`
  - `adios`
  - `gsl`
  - `fftw3`
  - `hypre`
  - `metis`
  - `mumps`
  - `netcdf`
  - `ocr`
  - `petsc`
  - `ptscotch`
  - `scalapack`
  - `superlu`
  - `trilinos`

7.2 Deprecated features and packages

The following features and packages are deprecated and will be removed in a future version of SUSE Linux Enterprise High Performance Computing.

- The "Hierarchical Data Format version 5" library and tools have been deprecated and will be removed from the HPC Module in SLES SP7. If you require this library beyond SLE HPC 15 SP6 (prerelease), use the Spack package manager to create it according to your requirements.
- `clustduct` is deprecated and will be removed in SUSE Linux Enterprise High Performance Computing 15 SP7. With SLE HPC 15 SP6 (prerelease), `warewulf4` has been introduced as cluster deployment tool, and users are advised to migrate to it.

8 Obtaining source code

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A Changelog for 15 SP6 (prerelease)

A.1 Pre-release

A.1.1 New

- Redefined release notes title and abstract to reflect module supplement status (*Article "SUSE Linux Enterprise High Performance Computing Release Notes (Supplement to the SUSE Linux Enterprise Server Release Notes)"*, Jira (https://jira.suse.com/browse/PED-7876)).

A.2 2025-10-31

A.2.1 New

- Added deprecation notice about `hdf5` in *Section 7.2, "Deprecated features and packages"* (Jira (https://jira.suse.com/browse/PED-12383))
- *Section 6.17.1, "Deprecation of old Versions of Slurm"* (Jira (https://jira.suse.com/browse/PED-6787))

A.3 2024-04-17

A.3.1 New

- *Section 6.1, "SLE HPC no longer a separate product"* (Jira (https://jira.suse.com/browse/PED-7684))

A.4 2023-10-19

- Initial SP6 release